Chapter 1

Applying Numerical Methods to Bivariate Data

1.1 Review of Scatter Plots

1.2 Correlation Coefficient

In the previous section, we used scatterplots to visually judge whether two variables are related. But how do we move beyond "this looks like a strong relationship" to something we can actually calculate and compare? That is exactly what the *correlation coefficient* gives us.

A correlation coefficient is a numerical measure used to judge the relation between two variables. The one we will use throughout this course is *Pearson's correlation coefficient*, also known simply as the correlation coefficient. It measures the strength and direction of the linear relation between two quantitative variables.

You will encounter two versions of this value depending on whether you are working with a full population or a sample drawn from one. When computed from a population, the correlation coefficient is written as ρ (read as "rho"). When computed from a sample, it is written as r. In this course, we will almost always be working with sample data, so r is the symbol you will use most.

No matter how strong or weak a relationship is, r always falls within a fixed range:

$$-1 \leq r \leq +1$$

Think about what those endpoints mean. A value of $r = +1$ or $r = -1$ means the points in a scatterplot fall perfectly on a straight line. A value of $r = 0$ means there is no linear relationship between the two variables at all. Every other value of r falls somewhere in between, and in the next subsection we will look at exactly how to interpret where r lands.

Exercise 1 Boundaries of r

Answer the following questions about the range of the correlation coefficient.

1. A student computes $r = 1.3$ for a dataset. What does this tell you, without looking at the data?
2. Two researchers study the same population but one uses the full population and the other uses a random sample. What symbol does each researcher use for their correlation coefficient?
3. Can r equal exactly 0.5? Can it equal exactly -1? Explain what each of those values would mean about the data.

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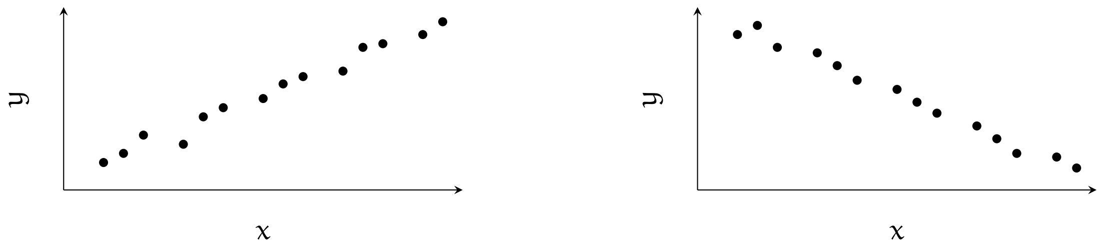
Direction

Now that we know r always falls between -1 and $+1$, the first thing to read from it is direction. The sign of r tells you immediately whether the two variables move together or against each other.

A positive value of r indicates that as x increases, y also tends to increase. Taller fathers tend to have taller sons, so the scatterplot shows an upward trend and r is positive.A negative value of r indicates that as x increases, y tends to decrease. As a phone gets older, its battery life tends to decrease, so the scatterplot shows a downward trend and r is negative.



(a) Positive relation ($r > 0$) (b) Negative relation ($r < 0$)

Figure 1.1: Direction of a linear relation as shown in a scatterplot.

Notice that direction says nothing about strength. Both $r = +0.1$ and $r = +0.95$ are positive, but they describe very different relationships. That is what we look at next.

Exercise 2 Reading Direction from r

For each pair of variables, predict whether r would be positive, negative, or zero.

x	y	Sign of r
Hours studied	Exam score	
Temperature	Hot drinks sold	
Shoe size	Intelligence	
Age of a car	Resale value	

Working Space

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Strength

Direction tells us which way a relationship goes, but not how strong it is. For that, we look at the numeric value of r, ignoring its sign. The further r is from zero, the stronger the linear relationship between the two variables. The closer r is to zero, the weaker it is.

At the extremes, $r = +1$ or $r = -1$ indicates a *perfect correlation*, meaning every point in the scatterplot falls exactly on a straight line. In practice, real data almost never achieves this. On the other end, $r = 0$ means there is no linear relationship at all.

To compare the strength of two correlations, just compare the absolute values. If $r = -0.86$ for one pair of variables and $r = 0.75$ for another, the first pair has the stronger linear relationship because $0.86 > 0.75$.

Since interpreting strength can be subjective, statisticians often use cutoff values as a rough guide. The diagram below shows one commonly used set of cutoffs. Keep in mind that these are conventions, not rules. The key principle is always the same: the further from zero, the stronger the correlation.

Very Strong	Strong	Weak	No Correlation	Weak	Strong	Very Strong
−1.00 to −0.85	−0.85 to −0.50	−0.50 to −0.10	−0.10 to +0.10	+0.10 to +0.50	+0.50 to +0.85	+0.85 to +1.00

Figure 1.2: One commonly used set of cutoff values for interpreting the strength of a correlation coefficient.

Exercise 3 Interpreting Strength

Working Space

The table below shows correlation coefficients computed from four different datasets. For each one, describe the direction and approximate strength of the relationship.

r	Direction	Strength
+0.92		
−0.34		
+0.07		
−0.88		

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Example

Let us put direction and strength together using a real dataset. Consider a sample of 10 father-son pairs. The heights (in inches) are recorded below.

Height of Father x (inches)	Height of Son y (inches)
66	66
66	65
67	67
67	70
68	67
68	70
69	70
70	72
71	72
73	74

Before computing anything, we check whether a correlation coefficient is appropriate. Correlation only applies to linear relationships, so we first make a scatterplot to confirm the relation between the two variables is linear.

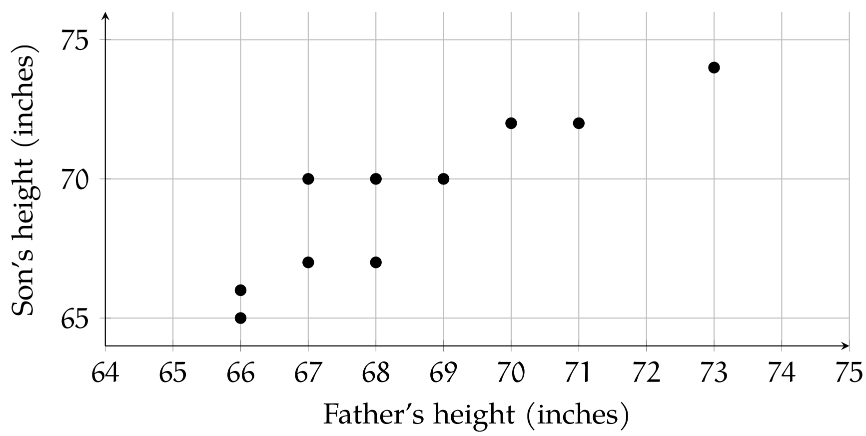

Figure 1.3: Scatterplot of father and son heights for a sample of 12 pairs.

The scatterplot shows a linear trend, so it is appropriate to compute r. In practice, correlation coefficients are computed using a calculator or statistical software rather than by hand. For this dataset, the computed value is approximately $r = 0.96$.

Computing r in Excel

Open your data in Excel or Google Sheets with the x values in column A and the y values in column B. To compute r directly, use the built-in correlation function:

```
=CORREL(A2:A11, B2:B11)
```

This returns the correlation coefficient between the two columns instantly. For the father-son data, this should return approximately 0.96.

Computing r in Python

```
import numpy as np

# Father and son heights in inches
x = [66, 66, 67, 67, 68, 68, 69, 70, 71, 73]
y = [66, 65, 67, 70, 67, 70, 70, 72, 72, 74]

r = np.corrcoef(x, y)[0, 1]
print(r)
```

`np.corrcoef()` returns a 2×2 matrix of correlation coefficients. The value at position `[0, 1]` is the correlation between x and y. For this dataset, the output should be approximately `0.96`.

Try replacing the data with your own values and observe how r changes as the relationship between the two variables becomes stronger or weaker.

What does this tell us? The sign is positive, so taller fathers tend to have taller sons. The value is 0.96, which falls in the very strong range of our cutoff table. The points cluster tightly around a straight line, and the scatterplot confirms this.

Notice also what r does not tell us. It describes the strength and direction of the linear relationship, but not the shape of any curve, and not whether one variable causes the other. A strong correlation between two variables does not mean one causes the other to change.

Exercise 4 Interpreting a Computed r

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sectionLinear Regression

Suppose you wanted to predict a teacher's starting salary based on how many years of experience they have. You might expect that more experience leads to a higher salary, but how do you turn that intuition into a precise estimate? That is exactly what a *linear regression model* does. It gives you a straight-line equation that describes the relationship between two variables, so you can use the value of one to predict the value of the other.

The Regression Model

When one variable depends on or is explained by another, we call the variable being predicted the *dependent variable*. The variable doing the explaining is the *independent variable*. In our example, starting salary is the response variable and years of experience is the independent variable, it would make little sense to say that a teacher's salary determines how many years they have worked.

The linear relation between two variables is described by the *population regression equation*:

$$Y = \alpha + \beta X$$

where Y is the response variable, X is the independent variable, α is the y-*intercept* (the value of Y when $X = 0$), and β is the *slope* (the amount Y changes for every one-unit increase in X). This equation describes the true relationship in the population, but in practice, we never know α and β exactly. We estimate them from sample data.

Once we have estimates for α and β, we can write the *estimated regression line*:

$$\hat{y} = a + bx$$

The symbol $\hat{y}$ (read "y-hat") is the *predicted value* of Y for a given value of X. Notice the difference in notation: α and β are population parameters; a and b are their sample-based estimates.

Now, no data will ever fall perfectly on a straight line. A student who studies ten hours might score differently than the line predicts. The difference between what the model predicts and what is actually observed is called the *residual*, or *error*, and is denoted e:

$$e = y - \hat{y}$$

A positive residual means the observed value was higher than predicted; a negative residual means it was lower. Going back to our teachers: if the model predicts a starting salary of \$38,000 for a teacher with four years of experience, but that teacher actually earns \$40,500, the residual is $40{,}500 - 38{,}000 = 2{,}500$. This teacher earned \$2,500 more than the model expected.

Exercise 5 Computing a Residual

A regression model predicts that a teacher with seven years of experience will have a starting salary of \$42,300. The teacher's actual starting salary is \$40,100.

1. Compute the residual.
2. Was the teacher's salary over-predicted or under-predicted? Explain.

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The goal of linear regression is to find the line that makes these residuals as small as possible across all observations.

The Least-Squares Regression Line

Every straight line you could draw through a scatterplot will produce a different set of residuals. Some lines will over-predict for certain point and under-predict for others. We want to know: which line does the best job overall? The answer is the *least-squares regression line*, also called the *line of best fit*. It is the line that minimizes the sum of the squared residuals, so, it makes $\sum e^2 = \sum (y - \hat{y})^2$ as small as possible.

Why should we square the residuals? First, squaring eliminates the sign, so positive and negative errors do not cancel each other out. Second, squaring penalizes large errors more heavily than small ones, which pushes the line toward a better overall fit. Now you know!

Exercise 6 Why Square the Residuals?

Suppose a regression line produces the following residuals for five data points: $3, -3, 2, -1, -1$.

1. Compute the sum of the residuals. What do you notice?
2. Compute the sum of the squared residuals.
3. Explain why simply minimizing the sum of the residuals would not be a useful goal.

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The least-squares regression line has two properties that always hold, regardless of the dataset.

First, it always passes through the point $(\bar{X}, \bar{Y})$ which represent the means (or average points) of the two variables. This point is sometimes called the *centroid* of the data.

Second, the slope b of the line is always given by:

$$b = r\left(\frac{S_y}{S_x}\right)$$

where r is the correlation coefficient, S_y is the standard deviation of the Y-values, and S_x is the standard deviation of the X-values. Notice that the slope depends directly on r: if there is no linear relationship ($r = 0$), the slope is zero, and the best prediction for any X is simply $\bar{Y}$.

Once you have b, the y-intercept a follows from the fact that the line passes through $(\bar{X}, \bar{Y})$:

$$a = \bar{Y} - b\bar{X}$$

These two formulas together give you the equation of the least-squares regression line:

$$\hat{y} = a + bx$$

Exercise 7 Computing the Line

For a dataset of teachers' years of experience (X) and starting salary in thousands of dollars (Y), the following summary statistics were computed: $\bar{X} = 6$, $\bar{Y} = 44$, $S_x = 3.2$, $S_y = 5.8$, $r = 0.91$.

1. Compute the slope b of the least-squares regression line.
2. Compute the y-intercept a.
3. Write the equation of the least-squares regression line.
4. The line must pass through the centroid of the data. Verify this using your equation.

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Interpreting Slope and Intercept

Having the equation of the least-squares regression line is only half the job. The other half is being able to explain what it means in plain language.

The slope b tells you how much the predicted value of Y changes for every one-unit increase in X. The key phrase you could see on an exam is **on average** because b is an estimate based on sample data, not an exact rule.

For our teachers example, if $b = 1.65$, the correct interpretation is:

> For every additional year of experience, the predicted starting salary increases by $1,650, on average.

Notice three things about that sentence. It names both variables. It gives the direction and amount of change. And it includes "on average." Leave out any one of those and the interpretation is incomplete.

Exercise 8 Interpreting the Slope

A least-squares regression line for predicting a teacher's starting salary (in thousands of dollars) from years of experience has slope $b = 1.49$.

1. Write a correct interpretation of the slope in context.
2. A student writes: "For every extra year of experience, salary goes up by \$1,490." What is wrong with this interpretation?

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The y-intercept a is the predicted value of Y when $X = 0$. Interpreting it requires care. Sometimes $X = 0$ is meaningful where a teacher with zero years of experience is simply a newly hired teacher. In that case the intercept has a natural interpretation: it is the predicted starting salary for someone entering the profession with no prior experience.

Other times $X = 0$ is outside the range of the data entirely, or physically impossible. In those situations, the intercept is mathematically necessary to define the line but has no useful real-world meaning. You should note that and move on.

Exercise 9 Interpreting the Intercept

Using the regression equation $\hat{y} = 34.10 + 1.649x$, where x is years of experience and $\hat{y}$ is predicted starting salary in thousands of dollars:

1. Write a correct interpretation of the y-intercept in context.
2. Is this interpretation meaningful here? Explain.

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Worked Example

A random sample of 10 teachers was selected from a large urban school district. Each teacher's years of experience at the time of hiring and their starting salary (in thousands of dollars) were recorded. The data is given in the table below.

Experience (years)	**Starting Salary ($1000s)**
3	36
8	43
1	32
0	30
5	38
10	47
2	33
7	42
4	37
6	41

Before doing any calculations, look at the data. As years of experience increase, starting salaries appear to rise steadily. There are no obvious outliers, and the values are spread across a reasonable range of experience levels. A scatterplot would confirm whether this relationship is linear and this is! There is a strong positive linear relationship between experience and starting salary.

From the data, we have $n = 10$ pairs of measurements. Using a calculator or software, we can obtain the following summary statistics:

$$\bar{X} = 4.6, \quad \bar{Y} = 37.9, \quad S_x = 3.169, \quad S_y = 5.013, \quad r = 0.9964$$

Exercise 10 Slope, Intercept, and the Regression Equation

Using the summary statistics above:

1. Compute the slope b of the least-squares regression line. Interpret it in context.
2. Compute the y-intercept a. Interpret it in context.
3. Write the equation of the least-squares regression line.

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Exercise 11 Prediction, Residuals, and R^2

Using the regression equation $\hat{y} = 30.650 + 1.576x$:

1. Predict the starting salary for a teacher with 5 years of experience.
2. The teacher in the sample with 8 years of experience has an actual starting salary of $43,000. Compute the residual and interpret it.
3. Compute the coefficient of determination R^2 and interpret it in context.
4. About what percentage of the variation in starting salaries is *not* explained by years of experience? Suggest one factor that might account for this.

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textbfComputing in Excel

You do not need to compute the slope, intercept, and R^2 by hand every time. Excel can produce all of these values at once, along with a scatterplot showing the line of best fit. Here is how.

First, enter your data into two columns, with experience values in column A and starting salary values in column B, with headers in row 1. Select both columns, then insert a scatterplot using **Insert** → **Chart** → **Scatter**. Once the chart appears, click on any data point, then select **Add Trendline**. In the trendline options panel on the right, choose **Linear** and check both **Display Equation on chart** and **Display R-squared value on chart**. Excel will draw the line of best fit and print the equation and R^2 directly on the chart.

To get the full regression output, use the **Data Analysis** Toolpak. Go to **Data** → **Data Analysis** → **Regression**. Set the Input Y Range to your salary column and the Input X Range to your experience column. Check **Labels** if your columns have headers. Click OK and Excel will produce a full regression table in a new sheet. The values you need are in the **Coefficients** column: the row labelled **Intercept** gives you a, and the row labelled

Experience gives you b. The R^2 value appears near the top of the output under **R Square**.

Computing in Python

Python can produce the same regression results as Excel, and a little more besides. The code below uses two libraries: `numpy` for the summary statistics and `scipy` for the regression, and `matplotlib` to draw the scatterplot with the line of best fit.

```
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

# Data
experience = [3, 8, 1, 0, 5, 10, 2, 7, 4, 6]
salary     = [36, 43, 32, 30, 38, 47, 33, 42, 37, 41]

# Least-squares regression
slope, intercept, r, p, se = stats.linregress(experience, salary)

print(f"Slope (b):     {slope:.3f}")
print(f"Intercept (a): {intercept:.3f}")
print(f"r:             {r:.4f}")
print(f"R-squared:     {r**2:.4f}")

# Scatterplot with regression line
x_line = np.linspace(0, 10, 100)
y_line = intercept + slope * x_line

plt.scatter(experience, salary, color="steelblue", label="Data")
plt.plot(x_line, y_line, color="firebrick",
         label=f"y = {intercept:.2f} + {slope:.2f}x")
plt.xlabel("Experience (years)")
plt.ylabel("Starting Salary ($1000s)")
plt.title("Experience vs. Starting Salary")
plt.legend()
plt.show()
```

Running this code produces the following output:

```
Slope (b):     1.673
Intercept (a): 30.203
r:             0.9957
R-squared:     0.9915
```

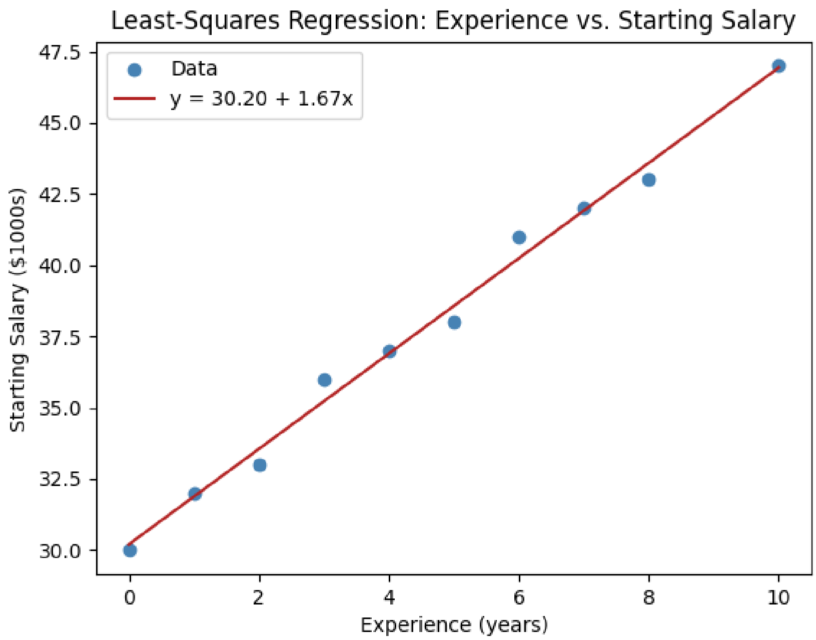

Figure 1.4: Experience vs. Starting Salary Graph generated from Python

These match the values we computed by hand exactly. The `stats.linregress` function does all the heavy lifting! It returns the slope, intercept, correlation coefficient r, a p-value, and the standard error of the slope in one line. For now, focus on the first four outputs. The p-value and standard error will become important when we study inference for regression later!

This is a draft chapter from the Kontinua Project. Please see our website (`https://kontinua.org/`) for more details.

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 2)

1. A value of $r = 1.3$ is impossible. The correlation coefficient must satisfy $-1 \leq r \leq +1$, so the student made a calculation error.
2. The researcher using the full population uses ρ. The researcher using a sample uses r.
3. Both are possible. An r of 0.5 would indicate a moderate positive linear relationship. The points follow an upward trend but with noticeable scatter. An r of -1 would indicate a perfect negative linear relationship, meaning all points fall exactly on a downward-sloping line.

Answer to Exercise 2 (on page 3)

x	y	Sign of r
Hours studied	Exam score	Positive
Temperature	Hot drinks sold	Negative
Shoe size	Intelligence	Zero
Age of a car	Resale value	Negative

Answer to Exercise 3 (on page 4)

r	Direction	Strength
$+0.92$	Positive	Very strong
-0.34	Negative	Weak
$+0.07$	Positive	No correlation
-0.88	Negative	Very strong

Answer to Exercise 4 (on page 6)

Answer to Exercise 5 (on page 8)

1. $e = 40{,}100 - 42{,}300 = -2{,}200$.
2. The salary was over-predicted. The observed salary was $2,200 less than the model expected.

Answer to Exercise 6 (on page 9)

1. $3 + (-3) + 2 + (-1) + (-1) = 0$. The positive and negative residuals cancel out, making the sum zero even though the line does not fit perfectly.
2. $3^2 + (-3)^2 + 2^2 + (-1)^2 + (-1)^2 = 9 + 9 + 4 + 1 + 1 = 24$.
3. A line that over-predicts some values and under-predicts others by equal amounts would have a residual sum of zero but it would be a poor fit. Squaring forces every error to contribute positively, so the minimization is meaningful!

Answer to Exercise 7 (on page 10)

1. $b = 0.91\left(\dfrac{5.8}{3.2}\right) = 0.91 \times 1.8125 \approx 1.649$
2. $a = 44 - 1.649(6) = 44 - 9.897 \approx 34.103$
3. $\hat{y} = 34.103 + 1.649x$
4. Substituting $x = \bar{X} = 6$: $\hat{y} = 34.103 + 1.649(6) = 34.103 + 9.897 = 44.000 = \bar{Y}$. ✓

Answer to Exercise 8 (on page 11)

1. For every additional year of teaching experience, the predicted starting salary increases by $1,490, on average.
2. The interpretation is missing the phrase "on average" and the word "predicted." The slope is an estimate from sample data, not a guarantee that every teacher's salary increases by exactly $1,490 per year.

Answer to Exercise 9 (on page 12)

1. The predicted starting salary for a teacher with zero years of experience is $34,100.
2. Yes, it is meaningful in this context. A teacher with no prior experience is a realistic case, so $X = 0$ is within the plausible range of the data.

Answer to Exercise 10 (on page 13)

1. $$b = r\left(\frac{S_y}{S_x}\right) = 0.9964\left(\frac{5.013}{3.169}\right) \approx 1.576$$

   For every additional year of teaching experience, the predicted starting salary increases by $1,576, on average.

2. $$a = \bar{Y} - b\bar{X} = 37.9 - 1.576(4.6) \approx 30.650$$

   The predicted starting salary for a teacher with no prior experience is $30,650. Since $X = 0$ represents a newly hired teacher with no experience, this interpretation is meaningful in context.

3. The equation of the least-squares regression line is:

   $$\hat{y} = 30.650 + 1.576x$$

   where x is years of experience and $\hat{y}$ is the predicted starting salary in thousands of dollars.

Answer to Exercise 11 (on page 14)

1. $$\hat{y} = 30.650 + 1.576(5) = 30.650 + 7.880 = 38.530$$

   The predicted starting salary for a teacher with 5 years of experience is $38,530.

2. $$\hat{y} = 30.650 + 1.576(8) = 30.650 + 12.608 = 43.258$$

   $$e = y - \hat{y} = 43 - 43.258 = -0.258$$

   The residual is −$258. The teacher's actual starting salary was $258 less than the model predicted. The salary was slightly over-predicted.

3. $$R^2 = r^2 = (0.9964)^2 \approx 0.9928$$

   About 99.28% of the variation in starting salaries is explained by the linear relationship with years of experience. This is an exceptionally strong fit.

4. About $100 - 99.28 = 0.72\%$ of the variation is unexplained. This is very small, but it may be due to factors such as the teacher's level of education, the subject area taught, or the specific school within the district.

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